

Arithmetic Sequences — nth Term

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

Method

Find the common difference d Rule starts with dn Constant = first term - d nth term = dn + constant

Check with $n = 1, 2, 3$

Common mistakes

- Writing $d + n$ instead of dn .
- Wrong sign on the constant.
- Confusing term number with term value.
- Not checking that n is a positive whole number.

Worked example

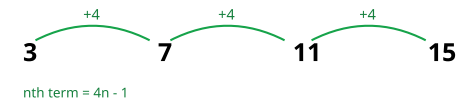
Find the nth term of 3, 7, 11, 15, ...

$$d = 4 \quad 4n; \text{ constant} = 3 - 4 = -1.$$

$$\text{nth term} = 4n - 1 \text{ (check } n = 1 \text{ gives 3)}$$

Visual aid

Visual Example



The arcs between terms show the constant difference (+4 each step). The rule $4n - 1$ gives the value of the term in position n .

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Write the next two terms: a) 2, 5, 8, 11, ... b) 20, 17, 14, ...

Q2 of 8

[2] ● ○ ○ ○ ○

Find d and the nth term: a) 4, 6, 8, 10, ... b) 1, 5, 9, 13, ...

Q3 of 8

[3] ● ● ○ ○ ○

Find the nth term: a) 5, 8, 11, 14, ... b) 7, 12, 17, 22, ... c) 10, 8, 6, 4, ...

Q4 of 8

[3] ● ● ○ ○ ○

nth term = $4n - 1$. a) 10th term? b) 20th term? c) Which term is 51?

Q5 of 8

[3] ● ● ● ○ ○

Is the number in the sequence? a) 100 in $3n + 1$? b) 50 in $4n - 2$? c) 30 in $5n + 2$?

Q6 of 8

[3] ● ● ● ○ ○

Theatre: row 1 = 15 seats, row 2 = 18, row 3 = 21. a) Rule for row n. b) Seats in row 10.

Q7 of 8

[4] ● ● ● ● ○

nth term = $6n - 4$. a) First three terms. b) 50th term. c) Show 201 is not a term.

Q8 of 8

[5] ● ● ● ● ●

A: $4n+1$, B: $3n+7$. a) First five terms of each. b) Common value? c) Solve $4n+1 = 3n+7$ and explain.